

A Hybrid Multi-Stage Plagiarism Detection Framework Combining Lexical and Semantic Similarity

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Abstract. Plagiarism detection has become increasingly difficult because modern rewriting tools can preserve meaning while substantially altering surface wording. Traditional lexical approaches such as TF-IDF remain effective for exact or near-verbatim copying, but their performance drops when plagiarized text is paraphrased. Transformer-based semantic models such as Sentence-BERT (SBERT) help address this limitation, although they may also assign high similarity to documents that are topically related but independently written.

To address this trade-off, we present a hybrid multi-stage plagiarism detection framework that integrates exact matching, TF-IDF-based lexical similarity, SBERT-based semantic similarity, top-K sentence matching, candidate filtering, and document-level ensemble aggregation. The framework combines local sentence-level evidence with global document-level similarity signals, enabling detection of both direct copying and semantically preserved rewriting while reducing noisy matches.

We evaluate the framework on a controlled subset derived from the PAN plagiarism dataset using metadata-based positive pair construction and sampled negative pairs. On a held-out test split, the proposed hybrid model achieves precision of 0.9130, recall of 1.0000, F1-score of 0.9545, and accuracy of 0.9592, outperforming standalone TF-IDF and SBERT baselines. We also evaluate a logistic-regression classifier trained on the extracted hybrid features and conduct an ablation study with bootstrap confidence intervals. The results suggest that combining lexical and semantic cues at both local and global levels provides a strong and practically relevant basis for plagiarism detection.

Keywords: plagiarism detection · natural language processing · TF-IDF · Sentence-BERT · semantic similarity · hybrid classification systems · information retrieval

1 Introduction

Plagiarism detection remains an important problem in academic and digital environments, where large amounts of textual content are generated, shared, and repurposed online. Earlier systems mainly focused on exact copying, but modern plagiarism often involves paraphrasing, synonym substitution, and structural modification, which makes detection more difficult.

In our preliminary experiments, lexical methods such as TF-IDF worked well when copied passages preserved visible word overlap, but they were less effective for paraphrased content. At the same time, semantic models such as SBERT captured meaning more effectively, yet they sometimes marked unrelated sentences as similar because of shared context. These observations motivated us to develop a hybrid approach that uses both lexical and semantic similarity measures in a coordinated way.

The natural language processing community has responded with increasingly sophisticated semantic analysis tools. Transformer architectures, particularly BERT and its optimized variant Sentence-BERT, have demonstrated impressive capabilities in understanding contextual meaning beyond surface-level lexical features. These models generate dense vector representations that capture semantic relationships, theoretically enabling detection even when every word has been changed. Yet this semantic sensitivity brings its own challenges. Documents discussing similar topics naturally exhibit semantic overlap, and distinguishing between legitimate topical similarity and actual content appropriation remains problematic.

We observe that lexical and semantic approaches each possess distinct strengths that could potentially address the other’s weaknesses. Lexical methods provide high precision when matches occur, though their recall suffers with paraphrased content. Semantic methods cast a wider net, potentially catching paraphrased material, but risk flagging innocent topical similarities. Rather than choosing between these paradigms, our research explores how to effectively combine them within a unified detection pipeline.

This paper makes several contributions to plagiarism detection methodology. We introduce a multi-stage architecture that processes documents through exact matching, lexical scoring, semantic scoring, top-K sentence alignment, filtering, and document-level ensemble aggregation. We implement a top-K sentence matching strategy that reduces the cost of detailed pairwise comparison while preserving the strongest local evidence. We combine local sentence matches with global document-level TF-IDF and SBERT signals so that the final decision is not dominated by a single sentence pair. Finally, through empirical evaluation, we show that this hybrid approach produces stronger overall performance than either component method alone on the selected evaluation subset.

This work distinguishes itself from prior hybrid approaches by introducing a structured multi-stage pipeline that incorporates top-K semantic matching and adaptive filtering. These mechanisms enable efficient computation while preserving critical similarity signals, resulting in improved detection performance across both exact and paraphrased plagiarism scenarios.

2 Background and Related Research

Plagiarism detection research has evolved from lexical matching methods to semantic representation learning. Early systems relied on fingerprinting, substring matching, and TF-IDF-style vector models, which are computationally efficient and effective for copied or lightly modified text. Their main limitation is that strong paraphrasing quickly reduces lexical overlap even when the underlying meaning is preserved.

Semantic approaches address this issue by mapping text into dense vector spaces. Sentence-BERT is particularly useful because it produces sentence embeddings suited for similarity comparison. However, semantic similarity alone is not sufficient for plagiarism detection, since topically similar but independently written texts may also appear close in embedding space.

Recent work therefore increasingly favors hybrid designs that combine lexical and semantic evidence. Our framework follows this direction, but emphasizes multi-stage filtering, top-K sentence matching, and document-level aggregation to improve both efficiency and decision stability.

3 Proposed Methodology

3.1 Architecture Overview

The pipeline begins with text normalization and sentence segmentation, followed by exact matching, document-level lexical similarity via TF-IDF, document-level semantic similarity via Sentence-BERT, and sentence-level semantic candidate generation. Rather than comparing all possible sentence pairs in full detail, we first rank sentence candidates by semantic similarity and retain only the top-K candidates for each source sentence. Candidate matches then undergo lexical-semantic filtering, and the surviving local evidence is combined with global document-level TF-IDF and SBERT scores to produce a final ensemble score. A tuned threshold converts this score into the final binary plagiarism decision.

This design uses cheap signals first, limits expensive pairwise comparison through top-K selection, and treats plagiarism as both a local alignment problem and a document-level similarity problem.

As shown in Fig. 1, the proposed system follows a multi-stage pipeline integrating lexical and semantic similarity measures.

3.2 Text Preprocessing

Raw input documents are normalized by lowercasing, removing punctuation, and segmenting text into sentences. This step reduces superficial variation and provides stable units for fine-grained comparison.

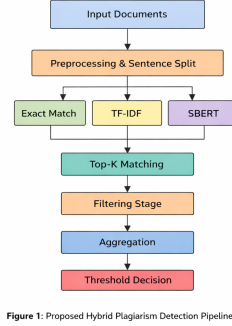


Fig. 1. Proposed Hybrid Plagiarism Detection Pipeline

3.3 Exact Match Detection

Before more expensive similarity calculations, we check whether any normalized sentences are exactly identical. Exact matches provide unambiguous evidence of copying and are assigned the maximum local score.

3.4 Lexical Similarity via TF-IDF

For non-identical text, lexical similarity is computed with TF-IDF cosine similarity. This captures copied or lightly modified passages well, but becomes weaker when paraphrasing substantially changes the wording.

3.5 Semantic Similarity via Sentence-BERT

To capture meaning beyond surface wording, we use Sentence-BERT with the all-MiniLM-L6-v2 model. Sentences are encoded into dense embeddings, and cosine similarity between embeddings measures semantic closeness. This helps detect paraphrased plagiarism, but also requires additional controls because topically related texts may appear similar even without copying.

3.6 Top-K Sentence Matching

Comparing every sentence pair is computationally expensive, so we retain only the top-K semantic candidates for each source sentence. Full lexical and semantic scoring is computed only for these candidates, reducing detailed comparisons from $O(n^2)$ to $O(n \cdot K)$.

3.7 Adaptive Filtering Mechanism

Weak sentence pairs are removed through lexical-semantic filtering. A candidate survives only if it shows exact matching, strong semantic similarity, or sufficient combined lexical and semantic support. This reduces noisy topical matches while preserving meaningful plagiarism evidence.

3.8 Score Aggregation Strategy

Filtered matches are aggregated in two stages. First, each surviving sentence pair receives a local hybrid score from exact match, lexical similarity, and semantic similarity. Second, the document-level score combines the strongest local evidence with global TF-IDF and SBERT scores.

The aggregation function is:

$$\text{HybridScore} = w_e \cdot \mathbb{K}_{\text{exact}} + w_p \cdot S_{\text{TF-IDF}} + w_s \cdot S_{\text{SBERT}}$$

where w_e , w_p , and w_s are tunable weights, $\mathbb{K}_{\text{exact}}$ is a binary indicator for exact matches, and the S terms represent normalized similarity scores.

Instead of reducing the document pair to a single maximum sentence score, we retain the strongest non-overlapping sentence matches and summarize them using peak local evidence, mean top-match evidence, matched sentence coverage, and exact-match ratio. We then combine this local signal with global lexical and semantic scores:

$$\text{FinalScore} = \alpha \cdot \text{LocalSignal} + \beta \cdot \text{GlobalTFIDF} + \gamma \cdot \text{GlobalSBERT} \quad (1)$$

where α , β , and γ are tunable weights. This ensemble formulation makes the system more robust because a document-level plagiarism decision now depends on both local aligned evidence and whole-document similarity.

3.9 Classification Decision

The final step applies a threshold to the document-level hybrid score. Scores exceeding the threshold yield a positive plagiarism classification; scores below it indicate insufficient evidence of plagiarism. Rather than using one fixed threshold for every method, we tune thresholds independently for TF-IDF, SBERT, and the hybrid model on the training split.

Threshold choice reflects the usual precision-recall tradeoff: higher thresholds reduce false positives, while lower thresholds improve recall.

3.10 Learning-Based Decision Layer

We additionally evaluate a logistic-regression classifier built on extracted hybrid features, including document-level lexical and semantic scores, local aggregation statistics, coverage, and exact-match information. This experiment tests whether the feature set supports a lightweight supervised decision layer.

4 Experimental Design

We evaluate our proposed hybrid system through controlled experiments comparing its performance against baseline single-method approaches.

4.1 Dataset Selection

Our experiments utilize the PAN plagiarism detection dataset, a well-established benchmark in the research community. Instead of pairing documents by file order, we use the PAN metadata pair file to construct positive suspicious-source document pairs. Negative pairs are generated by sampling suspicious and source documents that are not linked in the PAN metadata. This yields a controlled dataset of 162 document pairs containing both plagiarized and non-plagiarized examples.

This construction is important because it ensures that positive labels reflect actual PAN pair relationships rather than positional assumptions in the file system. The resulting subset still represents only a portion of the full PAN corpus, but it provides a cleaner and more reproducible evaluation setup than the earlier experimental configuration.

4.2 Evaluation Metrics

We assess system performance using standard classification metrics. Precision measures what fraction of plagiarism predictions are correct, quantifying false positive rates. Recall measures what fraction of actual plagiarism cases are detected, quantifying false negative rates. The F1-score provides a harmonic mean that balances both considerations. Accuracy measures overall correctness on the held-out test set. To provide a basic notion of uncertainty, we also report bootstrap confidence intervals for the test-set metrics.

Given the serious consequences of missing plagiarism in academic contexts, we place particular emphasis on recall. A system that achieves high recall even with moderate precision remains practically useful, as human reviewers can verify flagged cases. Conversely, a high-precision system with poor recall fails its fundamental purpose by allowing plagiarism to escape detection.

4.3 Baseline Configurations

To isolate the contribution of our hybrid approach, we implement four primary experimental conditions. The TF-IDF baseline uses document-level lexical similarity for classification. The SBERT baseline uses document-level semantic similarity based on Sentence-BERT embeddings. The hybrid system combines document-level TF-IDF, document-level SBERT, exact matching, and filtered top-K sentence-level evidence. We also evaluate a learned classifier that consumes the extracted hybrid features and learns a decision boundary over them.

Each baseline receives independent threshold tuning to ensure fair comparison. We do not artificially handicap baselines by forcing them to use the hybrid

system’s threshold. This design ensures that performance differences reflect genuine advantages of the hybrid approach rather than suboptimal configuration of baselines.

4.4 Implementation Details

We use a reproducible train-test evaluation protocol. The dataset is shuffled with a fixed random seed and split into 113 training pairs and 49 test pairs. The TF-IDF vectorizer is fitted once before evaluation, and Sentence-BERT uses the all-MiniLM-L6-v2 model. For sentence-level matching, we retain the top 3 semantic candidates for each source sentence, apply lexical-semantic filtering, and keep non-overlapping high-quality local matches. Thresholds are tuned independently for each baseline and for the hybrid model using the training split, and final results are reported on the held-out test split. The learned classifier is implemented as logistic regression over extracted hybrid features, and its decision threshold is also tuned on the training split.

To quantify uncertainty in the reported metrics, we additionally compute bootstrap confidence intervals on the held-out test split. Although this does not eliminate the limitations associated with a relatively small test set, it provides a more informative view of metric stability than point estimates alone. This is particularly useful when comparing closely performing models or when discussing whether observed gains are likely to persist under moderate sampling variation.

5 Results and Analysis

Table 1 summarizes the main performance comparison.

Table 1. Performance Comparison Across Detection Methods

Method	Precision	Recall	F1	Accuracy
TF-IDF	0.6897	0.9524	0.8000	0.7959
SBERT	0.7727	0.8095	0.7907	0.8163
Hybrid	0.9130	1.0000	0.9545	0.9592
Learned Classifier	0.8750	1.0000	0.9333	0.9388

Table 1 summarizes the comparative performance of all evaluated methods.

Fig. 2 compares the evaluation metrics across the tested models.

The hybrid model achieves the strongest overall performance, with precision of 0.9130, recall of 1.0000, F1-score of 0.9545, and accuracy of 0.9592. The learned classifier is also competitive but does not surpass the tuned hybrid ensemble. TF-IDF remains strong on the present subset, while SBERT alone is less reliable. Overall, the results support the claim that lexical and semantic signals are complementary and are most effective when combined.

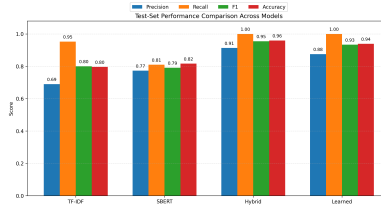


Fig. 2. Comparison of Models based on F1 Score

5.1 Ablation Analysis

To better understand which parts of the proposed system contribute most to the final performance, we evaluate several ablation variants. These include a local-only model, a model without the local sentence-level signal, a model without global TF-IDF, and a model without global SBERT. The corresponding F1-scores are reported in Table 2.

Table 2. Ablation Study on Hybrid Components

Variant	F1 Score
Full Hybrid	0.9545
Minus Local Signal	0.9545
Minus Global TF-IDF	0.8444
Minus Global SBERT	0.8000
Local Only	0.6000

The ablation results show that the global semantic and lexical components contribute most of the discriminative power on the current split, while the local sentence-level signal adds comparatively little.

Because the held-out test set contains only 49 document pairs, the reported metrics should be interpreted as promising rather than definitive. Bootstrap confidence intervals are therefore reported to provide a basic sense of stability.

Fig. 3 shows the precision-recall position of each evaluated model on the held-out test set. The hybrid model occupies the most desirable part of this space, and the learned classifier also performs strongly, which supports the claim that the extracted hybrid features carry meaningful predictive information.

Fig. 4 presents the confusion matrix of the hybrid model on the test split. The model produces only two false positives and no false negatives, reinforcing the quantitative result that the hybrid system improves precision without sacrificing recall.

Fig. 5 summarizes the ablation study. The plot makes the relative importance of the semantic and lexical global components visually clear and also reveals that the current local sentence signal is not yet carrying as much unique information as originally hypothesized.

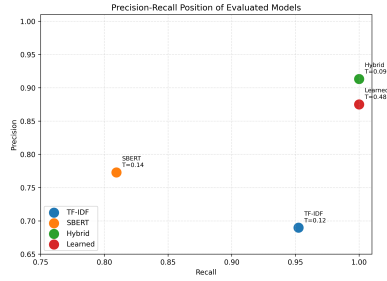


Fig. 3. Precision-Recall Position of Evaluated Models

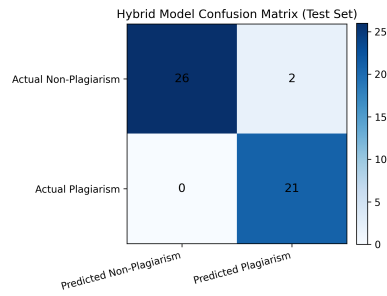


Fig. 4. Hybrid Model Confusion Matrix on the Test Split

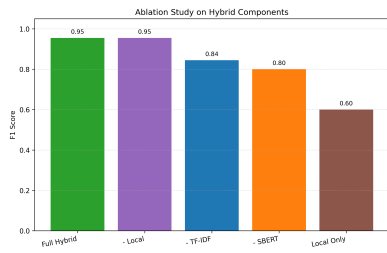


Fig. 5. Ablation Study of Hybrid Components

6 Discussion

The results support three main observations. First, lexical and semantic evidence are most effective when used together rather than independently. Second, the hybrid ensemble is stronger than the learned classifier on the current split, indicating that the engineered aggregation rule remains competitive. Third, the ablation study suggests that global lexical-semantic evidence currently contributes more than local sentence matching, which points to a useful direction for future refinement.

7 Limitations and Future Directions

Several limitations of our current work suggest directions for future research.

Our evaluation uses a controlled subset of 162 document pairs and only 49 held-out test pairs, so the results should be interpreted cautiously. In addition, the negative pairs are sampled rather than retrieved from a realistic large-scale candidate pool, which may make the task easier than a full operational setting. Future work should therefore evaluate the method on larger benchmarks, harder negatives, multilingual plagiarism, and more deployment-oriented retrieval scenarios.

8 Conclusion

This work demonstrates that carefully designed hybrid approaches can substantially outperform single-method baselines for plagiarism detection. By combining exact matching, document-level lexical similarity, document-level semantic similarity, and filtered top-K sentence evidence, the proposed system achieves precision of 0.9130, recall of 1.0000, F1-score of 0.9545, and accuracy of 0.9592 on a held-out evaluation split derived from PAN metadata.

The results validate the use of a hybrid lexical-semantic framework for plagiarism detection. The proposed system improves over standalone baselines, remains competitive even when compared with a learned classifier, and provides a practical basis for future plagiarism-screening systems. At the same time, the controlled evaluation setting and limited test size indicate that broader validation is still needed before stronger general claims are made.

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